

Review of Knee Proprioception and the Relation to Extremity Function After an Anterior Cruciate Ligament Rupture

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Several tests of human conscious knee proprioception have been described, but there is no consensus or reference standard established. Difficulties remain in the separation of information originating from muscles, tendons, and joints, and the tests cannot discriminate between loss of afferent signals or altered activity in the remaining receptors. There is convincing evidence from several descriptive studies that the afferent information is altered after a knee ligament injury and severely disturbed in some patients with anterior cruciate ligament (ACL) injuries. However, an inherent inferior proprioceptive ability may also exist in some individuals, which makes them vulnerable to injuries. The deficits in proprioception have mostly been studied and related to the consciously registered sense, whereas the extent of possible disturbances of the unconscious or reflexory mechanisms is largely unknown. The latter may, at least from a theoretical point of view, be predominantly contributing to the overall afferent regulation, and a possibility for major defects thus exists, since there is no knowledge of the quantified relation between the conscious and unconscious part. The clinical importance of the altered afferent information has not been evaluated properly, and the role of proprioception that contributes to function has yet to be investigated. A higher physiological sensitivity to detecting a passive joint motion closer to full extension has been found both experimentally and clinically, which may protect the joint due to the close proximity to the limit of joint motion. Proprioception has been found to have a relation to subjective knee function, and patients with symptomatic ACL deficiency seem to have larger deficits than asymptomatic individuals. Little is known about whether training can restore defects in sensory information or by which mechanisms possible compensatory pathways are established. In rehabilitation, each patient must, however, create muscle strength, alertness, and stiffness in harmony with the disturbed mechanics of the knee, which are present both after nonoperative treatment of the ACL and after a reconstruction of the ACL. *J Orthop Sports Phys Ther* 2001;31:567-576.

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An injury to the anterior cruciate ligament (ACL) results in an increased *envelope of motion*.¹⁹ In patients with ACL insufficiency, large variations in instability and impairment have been reported, without any correlation to the increased laxity.^{2,11,26,60} Several studies have shown decreased proprioception of varying degrees after an ACL injury, as well as a correlation between the measured proprioceptive ability and the patient's subjective extremity function.^{11,39} The *envelope of function*, a concept introduced by Dye,³¹ is a load and frequency distribution that defines a safe range of loading for a given joint. The area within the envelope is the zone of "homeostatic loading," and loads beyond the margin of the envelope are in the zones of supraphysiological overload or structural failure. Loads that induce structural failure of a component, such as rupture of the ACL, decrease the future tolerance for a given load (ie, decrease the size of the envelope of function with a concomitant in-

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crease in the envelope of motion). The neuromuscular system consists of afferent information, central processing, and efferent activity with a muscular response. The aim of rehabilitation after an ACL injury, treated with or without surgery, is to improve and optimize sensorimotor control to obtain the largest possible envelope for homeostatic loading. This clinical commentary focuses on afferent information, possible relations to more complex functions, and different findings in symptomatic versus asymptomatic patients.

PROPRIOCEPTION IN THE HUMAN KNEE JOINT

Mechanoreceptors and Neural Pathways

The first histological demonstration of mechanoreceptors in the human ACL was reported by Schultz et al.¹⁰⁰ Zimny et al¹¹⁷ and Schutte et al¹⁰¹ verified the presence of these receptors and calculated that the neural elements constituted 1.0–2.5% of the tissue in the ACL. Subsequent studies by Haus and Halata,⁴⁹ Katonis et al,⁶³ and Krauspe et al⁶⁴ have described these receptors, but with somewhat different characteristics. The differences are probably caused by the controversy about the classification of the mechanoreceptors.⁵¹

The same receptors (Ruffini and Pacinian corpuscles, Golgi tendon organs, and free nerve endings) have also been identified in the human medial meniscus. Albright et al¹ localized the neural elements in the outer and middle thirds of the meniscal cartilage, but not in the nonvascularized inner third. The highest concentration was found in the meniscal horns, in particular the posterior horn. Assimakopoulos et al³ reported similar results. Furthermore, Haus and Halata⁴⁹ identified receptors in the human knee joint capsule, and De Avila et al²⁷ identified mechanoreceptors, classified only as “nonPaciniform endings,” in the human lateral collateral ligament.

Mechanoreceptors can be divided into rapidly and slowly adapting receptors. The Pacinian corpuscles are rapidly adapting receptors that decrease their discharge rate to extinction within milliseconds of the onset of a continuous stimulus; their main function is sensing sudden movement, acceleration, or deceleration at onset or termination of movement. The Ruffini ending and Golgi tendon organ, on the other hand, are slowly adapting, which means that they continue their discharge in response to a continuous stimulus, signal actual joint position, and slow movements.

Gomez-Barrena et al⁴³ and Parsch et al⁸⁸ have demonstrated direct neural pathways from the ACL to spinal ganglia by the use of tracer technique and axonal transportation. Together with visual and vestibular input, the input from mechanoreceptors in skin, muscles, tendons, ligaments, and joint capsules pro-

vide the central nervous system with information about limb position and enable postural control. The relative contribution of each anatomic receptor population is, however, still unclear. The proprioceptive input is carried along large myelinated axons with high conduction velocities (in contrast to the small unmyelinated pain fibers) to the dorsal root spinal ganglion. Some of the information then further ascends in the posterior column of the spinal cord to the post central gyrus of the cerebral cortex, which accounts for conscious awareness of position and movement characteristics of the extremities.⁵⁹ Sensory-evoked potential from electrical stimulation of the intact ACL can be registered in the cerebral cortex,⁹¹ and in a more recent study, alterations in central nervous system activity were registered after an ACL rupture as well as after an ACL reconstruction.¹¹⁰

Tests of Human Conscious Proprioception

Several tests for knee proprioception have been described, but no consensus or criterion reference has been established. Many testing procedures involve neutralization of visual, auditory, and cutaneous inputs and the occurrence of a random delay before onset of the movement to eliminate any external cues. However, difficulties remain in separating information originating from muscles, tendons, and joints.⁶ Also, these tests cannot discriminate between loss of afferent signals or altered activity in the remaining receptors.⁵¹ In patients with a unilateral knee disorder, the contralateral uninvolved knee may serve as an internal control, while nonimpaired knees from a reference population may serve as external controls.^{40,41,67,96} However, the possibility of a bilateral altered proprioception after a unilateral injury and methodological differences in patient positioning, starting position of the knee, and testing velocities make some of the results difficult to interpret.

The most established and probably the most sensitive test to clinically measure conscious proprioception is the threshold to detection of passive motion (TTDPM).^{5-9,17,20,23-25,33,38-41,46,47,68,69,71,73,85,86,94,96-99,102,104-106} This means that the knee is passively moved either into extension or flexion, and the subject responds as soon as he or she feels the movement. It has been suggested that joint receptors register movement but are unlikely to play any significant role for position sense.^{78,92} The TTDPM, tested at slow angular velocities (0.5–2.5°/s), is believed to maximally stimulate joint receptors and minimally stimulate muscle receptors;⁶⁷ however, this has not been experimentally proven.

Another common test, although with questionable accuracy and reliability,^{40,41,45} is reproduction of passive positioning (RPP). This test is performed by active reproduction of an angle to which the knee previously

has been positioned,^{6,8-10,18,25,30,33,40,41,44,48,62,96,97,104-106,109} although passive reproduction has also been described.^{24,30,71,89,99} Active reproduction of active motion has also been used,^{44,45,54,74,75,83} and another method is to reproduce a passive motion visually on a goniometer.^{11,12,22,40,41,56,57,96,97,112}

Other tests include the ability to actively reproduce or track a joint position or passive movement with the contralateral leg, a test sometimes called relative proprioception,^{24,62,79,80} and, finally, the ability to use leg motions to follow an oscillating (expanding to represent extension and contracting to represent flexion) bar-graphed video display has been studied on astronauts.¹⁶ Some of these repositioning tests have, however, been used without any theoretical reasoning as to their potential advantages and shortcomings as a measuring technique of "knee proprioception," and they also, per definition, include central processing and efferent activity.

Proprioception in Athletes Versus Controls

Proprioception in athletes has been investigated on ballet dancers^{5,8,10} and gymnasts.⁶⁸ These athletes use the extremes of flexibility, coordination, postural control, and balance during performance and were, therefore, logically supposed to have better proprioception than control groups.

Barrack et al studied proprioception in members of the New Orleans Ballet during 3 studies by using TTDPM^{5,8} and active RPP.^{8,10} Compared to an age-matched control group, the ballet dancers had significantly lower threshold values. On the other hand, the ballet dancers performed significantly worse than the control group on the active RPP test, with a tendency to overshoot in their reproductions. All ballet dancers participating in the above-mentioned studies were hypermobile. However, conflicting results in women with hypermobility have been reported in another study.⁴⁷ In that study, 10 women with hypermobility syndrome had significantly higher threshold values than control subjects with stiffer knees. The difference may, however, be due to methodological differences, as well as to the different levels of athletic training between the ballet dancers and the women with hypermobility syndrome.⁴⁷

Another study of proprioception in athletes measured TTDPM among women gymnasts.⁶⁸ The study compared 15 intercollegiate gymnasts and 30 non-gymnast controls; the gymnasts were found to have significantly lower threshold values than the control group.

Influence of Exercise on Proprioception

Exercise can be classified as fatiguing and nonfatiguing. The effects of both types of exercise on proprioception have been investigated. Skinner et al¹⁰⁶

showed a decline in proprioception (TTDPM and active RPP) after fatiguing exercises by 11 US Navy SEAL members with a high level of physical training, and they suggested a primary role for muscle receptors as a determinant of proprioception and a possible secondary role for joint receptors. Marks⁷⁴ had similar findings, although using active reproduction of active positioning, when studying postfatigue proprioception in young women.

In contrast, Stillman et al¹⁰⁹ studied the effect of nonfatiguing muscle exercise on proprioception in 20 elite men participants in Australian Rules football. Compared to an identical control group of 20 subjects, no significant differences were seen in proprioception as measured by active RPP, which was slightly modified from other studies (after passive positioning, subjects held their knees isometrically in that position for 5 seconds before the examiner returned the leg to the starting position). Thus, it seems that fatiguing, as opposed to nonfatiguing exercise, may reduce the sensitivity of the muscle spindles with a resultant decrease in joint position sense and movement sense.

Proprioception in Cruciate Ligament Deficiency

There are 2 studies examining proprioception after an isolated posterior cruciate ligament rupture^{23,99} which showed a significant decline in proprioception measured as TTDPM in the posterior cruciate ligament-deficient knees compared to nonimpaired knees. In the most recent study, passive RPP was also studied, and this generated conflicting results: posterior cruciate ligament-deficient knees performed better in RPP than nonimpaired knees from 110° but not from 45°, and worse in TTDPM at 45° but not from 110°.

There are many studies examining proprioception in patients with ACL injuries. The first study on proprioception in patients with ACL deficiency was made by Barrack et al,⁷ and significantly higher threshold values in TTDPM were found in ACL-deficient knees than in corresponding control knees. Their results have been confirmed by several studies using TTDPM,^{17,20,25,33,38-41,73,86,87,97,110} and there is, to our knowledge, only 1 study using TTDPM that did not show any proprioceptive deficits in ACL-deficient knees.¹¹⁴

Furthermore, it seems that symptomatic patients with ACL deficiency have significantly higher threshold values in TTDPM than asymptomatic patients.⁹⁷ There is deficient proprioception associated with ACL deficiency, meniscal and chondral lesions, and subjective knee function.³⁹ The results of 2 independent studies using TTDPM suggest that the knee is more sensitive to movements in the end range of motion (15-20°), in particular moving into extension, and that defects after an ACL rupture are best

detected or predominant in this flexion range.^{40,69} Moreover, some studies have shown that the contralateral nonimpaired knees serving as internal control knees had significantly inferior threshold values than external control knees in nonimpaired individuals, suggesting that bilateral defects may follow a unilateral rupture of the ACL. The contralateral knee of an impaired individual may consequently not serve as a reliable control.^{56,96}

An impaired proprioception in ACL deficiency has also been described in studies using visual estimation of passive motion on a goniometer.^{11,22,56} When it comes to RPP, some studies report inferior proprioception with ACL deficiency,^{25,33} whereas other studies have not found any differences between ACL-deficient knees and an age-matched nonimpaired control group, neither in visual estimation nor in RPP.^{40,41,97}

Finally, a study using the somewhat different method of active reproduction of passive and active motion in an upright patient with the tested leg hanging freely could not show any differences between the impaired and nonimpaired leg.⁴⁴

To sum up the knowledge about ACL deficiency and conscious proprioception, it seems that there are deficits after ACL rupture and that the TTDPM probably is the most sensitive method to prove these subtle changes. The effect of afferent regulation on the subconscious aspects of knee proprioception is more difficult to study, and possible defects in humans after a knee ligament injury are, to our knowledge, not known.

Effect of Rehabilitation on Proprioception in Anterior Cruciate Ligament Deficiency

The effect of a 4-week rehabilitation program that included proprioceptive training in patients with ACL deficiency has been investigated in 2 independent studies, one measuring proprioception as TTDPM and RPP,⁹⁰ the other as visual estimation of an angle change on a goniometer.²² After completion of the rehabilitation program, no improvement in proprioception was found in either study. In a prospective study⁴¹ on recent knee injuries, proprioception was measured repeatedly by TTDPM during a neuromuscular rehabilitation program. The goal was to improve sensorimotor control and functional stabilization of the limb by training coordination, postural reactions, and endurance.¹¹⁵ Initial proprioceptive defects tended to decrease with time, but whether this was due to the training requires further studies. These 3 studies indicate, however, that a 4-week rehabilitation program is not long enough to restore proprioception.

Anterior Cruciate Ligament Reconstruction and Proprioception

After the findings of impaired proprioception in patients with a ruptured anterior cruciate ligament, the next concern was to find out if ligament reconstruction could improve or restore the proprioceptive ability, but these studies are controversial. Theoretically, repairing the injured ligament, rather than substituting it with a graft, has the possibility of maintaining sensory properties.⁵⁹ However, clinical studies have unequivocally shown inferior clinical results with primary repair (suturing) alone compared to augmented repair or reconstruction with the use of a graft.³⁵ Moreover, studying human grafts histologically has produced conflicting results. One study found no signs of reinnervation in postoperative biopsies of patellar tendon grafts in 10 patients who had undergone ACL reconstruction;⁴ however, another study identified mechanoreceptors in 2 asymptomatic patients, but with failed ACL reconstructions 9 and 10 years after surgery.²⁹ Both patients in that study had a nonfunctional semitendinosus tendon graft, and it was suggested that reconstruction using a graft could maintain or develop proprioceptive properties. This is, however, not supported in the neurophysiological literature.⁵¹

The first study on proprioception and ACL reconstruction actually preceded the first study on proprioception and ACL deficiency. Newberg⁸³ studied the effect of ACL reconstruction on proprioception measured as active reproduction of active motion at flexion angles of 30° and 70° in 35 patients with ACL reconstruction and 20 nonimpaired control subjects. The patients with reconstruction performed significantly worse at 30° with the reconstructed knee than the nonimpaired knee. However, no significant difference was seen in the patient group at 70° or between the controls and the subjects with reconstruction at either angle.

Several TTDPM studies describe proprioceptive deficits after ACL reconstruction.^{33,69,96,110} One of these studies investigated TTDPM from starting positions of 15° and 45° in 12 patients with arthroscopic patellar tendon grafts. That study reported significant deficits in the reconstructed knee compared to the nonimpaired knee at 15° moving both into extension and flexion, but no difference at 45°.⁶⁹ Kinesthetic awareness was significantly more sensitive at 15° moving in both directions than at 45°, and similar findings have been reported in studies on ACL deficiency.^{20,40} Another study found, in addition to persisting proprioceptive deficits in ACL-reconstructed knees, significantly higher threshold values in uninjured knees of the patient group compared to uninjured knees in a nonimpaired age-matched control group, indicating bilateral proprioceptive defects.⁹⁶

One study using TTDPM reported lower threshold

values in reconstructed knees compared to knees of a control group, but these were still significantly worse than contralateral uninjured knees.²⁴ Yet another TTDPM study⁷³ showed no proprioceptive differences after ACL reconstruction. In that study, hamstrings and patellar tendon grafts were also compared, but no significant differences were found, indicating that the choice of surgical technique is of minor importance in this respect and as evaluated with the chosen measuring technique.⁷³

Furthermore, 2 studies using the visual estimation test,^{11,56} as well as a study using active RPP,³³ found impaired proprioception in reconstructed knees compared to knees of a control group; however, another 3 studies using active RPP found no differences between reconstructed knees and contralateral uninjured knees.^{30,48,96} One of those studies also evaluated passive RPP and found significantly lower error amplitudes in the reconstructed knees compared to the contralateral uninjured knees.³⁰ Thus, divergent findings about proprioception after ACL reconstruction have been reported in studies that used different measuring techniques.

Bracing and Proprioception

Bracing and bandaging have been widely used in sports medicine for decades, although the scientific basis for their use has been essentially unknown. During the last few years, however, bracing and bandaging have been evaluated in an increasing number of studies, including studies on proprioception.^{12,17,18,56,57,69,80,89,94}

Even if the reason is prophylactic or therapeutic, bracing and bandaging are used to support the joint mechanically. Traditionally, cutaneous afferents have been thought to contribute in a minor way to the proprioceptive information,⁷⁰ but recently it has been thought that bracing and bandaging may also affect proprioception by stimulating cutaneous receptors around the joint, whereas the muscle and joint receptors are probably situated too deep to be significantly affected.⁸⁹ Some studies show a positive effect on proprioception from wearing a brace or a bandage,^{12,18,56,57,69,80,89} and other studies show no effect at all.^{17,56,94}

DISCUSSION

Some reported defects in neuromuscular performance may have a relation to altered afferent information after a knee ligament injury and must, therefore, be considered in the treatment.

Quadriceps Strength Deficits

A persistent muscle atrophy of the quadriceps is a common finding, sometimes despite intense and reg-

ular training, and even if muscle volume is regained, muscular strength may remain reduced. A suboptimal activation of normally functioning muscle fibers has been suggested,³² and there may be a relation between the defects in muscle performance and the altered afferent information.

Joint Congruency (Anterior Tibial Translation)

Some patients with ACL deficiency have an inferior ability to keep the ACL-injured joint congruent during activities when quadriceps activity is unavoidable, such as climbing stairs¹¹¹ or during single limb stance in slight flexion.³⁶ In the latter study, symptomatic patients with insecurity and functional instability showed larger displacements with a measurable incongruency of the joint when compared with asymptomatic patients.

In the electrogoniometer study by Vergis,¹¹¹ an abnormal tibiofemoral relation was registered, with an anterior shift of the tibial resting position in relation to the femur in both ACL-deficient knees and after an ACL reconstruction. These patients were, however, able to resist further anterior tibial translation while walking stairs, but nothing was mentioned about whether they were symptomatic or not. The same author also described defects in the uninjured extremity, which may accord with the findings of an inferior proprioception also on the uninjured side.

Kvist⁶⁶ used the same technique, but with simultaneous electromyographic recordings, and found an anterior tibial translation in weight-bearing activities despite increased hamstrings activity. The study suggested that other factors, such as joint compressive forces and joint geometry, especially the posterior tilting of the tibia plateau, may be more important to resist anterior tibial translation in ACL deficiency. The study found that hamstrings co-contraction could not limit anterior tibial translation during weight-bearing, which further emphasizes the importance of and need for alternative agonistic ACL activity, both mechanically and as a sensor to regulate quadriceps force.

Muscle Recruitment and Reaction Time

An experimental study by Johansson et al⁵⁸ found a response in the hamstrings and the gastrocnemius to a load on the ACL. Solomonow et al¹⁰⁸ found that stress on the ACL inhibits the quadriceps and stimulates the hamstrings and that this reflex was slower in ACL-deficient knees, implying that a secondary reflex was initiated, maybe by capsular or other ligamentous receptors.

Beard et al¹³ found an increased reflex hamstrings contraction latency with a correlation to episodes of giving way in 30 patients with ACL deficiency compared to the uninjured contralateral knee. However,

a subsequent study of 11 patients with ACL deficiency failed to verify this latency,⁵⁵ whereas delayed muscle activation after an ACL injury was shown by Hopf et al.⁵²

Huston and Wojtyś⁵³ reported gender-related differences in muscle recruitment order. The same authors¹¹³ also found a superior ability to keep the joint congruent in asymptomatic versus symptomatic ACL-deficient knees during a weight-bearing stress test; and the asymptomatic patients preferred an initial recruitment of the hamstrings, while the symptomatic patients recruited the quadriceps first during voluntary response. They also found that acute ACL-injured extremities used a different muscle recruitment order than did chronic and normal extremities, thus indicating a learning effect or adaptation to the injury with its new envelope of motion. It was also found that voluntary muscle activity in the quadriceps, hamstrings, and gastrocnemius in the injured extremity was slower than on the uninjured side and in controls and that the uninjured side was slower than controls in activation of the quadriceps and the hamstrings. These authors concluded that, "Muscle recruitment patterns in response to anterior tibial stress appear to change with time from injury and the seemingly unwanted quadriceps response may be a result of postinjury dysfunction caused by the loss of protective afferent input previously originating in the ACL."¹¹³ This may agree with experimental EMG findings, which showed that EMG during static and dynamic shear forces applied to the anteromedial bundle of the ACL resulted in a significant increased hamstrings activity and decreased quadriceps activity.⁹³ The same principal findings have been recorded for EMG during walking⁷² and in stance phase,²¹ with an increased activity in the hamstrings and decreased in the quadriceps. A different motion pattern has been found to persist both after an ACL injury⁴² and after an ACL reconstruction.^{107,111}

Postural Control

The importance of proprioception for postural control is best exemplified when a patient with congenital loss of peripheral sensory information immediately falls to the floor when it suddenly becomes dark.⁸²

Patients with both chronic and symptomatic unilateral ACL deficiency were shown to have a bilateral defect in postural control when body sway was measured during the single limb stance on a force plate.¹¹⁶ After 3 months of neuromuscular rehabilitation, their uninjured side did not deviate significantly from a reference group, whereas the injured side still had an abnormal sway speed. At 12 months, no differences were found between the patients and the reference group, which was also the result at a follow-up after 3 years.¹¹⁶

The same principal findings were reported by Henriksson,⁵⁰ who reported that patients with chronic but asymptomatic ACL deficiency had impaired postural control, when measured in single limb stance on a force plate. A longer reaction time and greater sway amplitude in response to perturbations were also registered, and both the injured and uninjured extremities were affected.

After an ACL reconstruction and a neuromuscular rehabilitation program focused on range of motion, strength, agility, and postural control, there was no correlation between measured laxity and the ability to control perturbations. It seems that neuromuscular function plays an important role in postural control, and a positive effect from perturbation training has been reported.³⁴

Gait Analysis

During gait analysis, Berchuck et al¹⁵ recorded a quadriceps avoidance gait, characterized by a flexed knee gait and an absent extension moment, also affecting the uninjured extremity. However, not all patients with ACL injuries showed a quadriceps avoidance, which may agree with the findings that there is less anterior tibial translation in asymptomatic patients during single limb stance in slight flexion³⁷ despite their ACL deficiency. These patients were, thus, able to compensate for the increased laxity without having to inhibit the quadriceps, whereas symptomatic patients, when forced to use the quadriceps, could not keep their joint congruent. It is not clearly established whether the defects emanate from loss of information from ACL mechanoreceptors, from induced changes in the stimulation of remaining receptors due to the altered kinetics,⁵¹ or from defects in central processing or in the muscle response, and there may be an effect on more than one of these levels.

The flexed knee gait¹⁵ has been recorded by other investigators,^{14,61,84,95} and a relation to time-from-injury has been suggested.⁸⁴ Simultaneous EMG recordings have confirmed increased hamstrings activity in stance¹⁴ and prolonged duration of co-contractions in the quadriceps and hamstrings.⁹⁵ A relationship between quadriceps strength and gait parameters has also been found,¹⁰⁷ and in 2 studies^{15,95} a disturbed gait on the uninjured side was also reported.

Mechanical Factors

There are, however, also mechanical factors that may influence the patients' extremity function in a weight-bearing situation after a knee ligament injury. These factors may vary among individuals due to both a normal biological variation and the total extent of the knee injury (ie, the altered envelope of motion). Posterior tibia plateau tilting has been

shown to correlate with anterior tibial translation during weight-bearing,²⁸ and it has been suggested that the shape of the lateral tibia plateau would influence the magnitude of pivot shift^{65,77} and disability.⁶⁵ It has also been found that there is a biological variation in the shape of the femoral condyles and that symptomatic patients have a more spherical shape than patients with no major symptoms of instability after their ACL rupture.³⁶ These findings mimic the act of standing on a ball compared to the act of standing on a sled. In a dynamic situation with higher stresses, there is an increased demand on both the afferent and efferent part of the neuromuscular support, which, thus, may also be aggravated by these mechanical factors.

IMPLICATIONS OF RESEARCH

A combination of scientifically-evidenced defects in sensory function and characteristics of symptomatic and asymptomatic patients with ACL deficiency suggests that mechanoreceptors in the ACL have exclusive, predominant function as the primary antagonist to quadriceps muscle force near full extension when the hamstrings have an unfavorable force vector on anterior tibial translation. This sensory function is, thus, specialized to evaluate and regulate the quadriceps force. The strength deficits in the quadriceps after an ACL rupture may be an adaptation to the loss of facilitating signals that report strain within harmless limits (ie, within the area of homeostasis). The reported defects in proprioception near full knee extension (the segment of the range of motion with the highest physiological sensitivity for joint motions) may result from the loss of specific ACL receptors. The flexed knee gait often reported after ACL injury may be an adaptation designed to avoid a dangerous zone near the border of the envelope of motion.

Based on this assumption, all of the following may be explained: persistent quadriceps atrophy, suboptimal activation of normally functioning quadriceps muscle fibers, the quadriceps avoidance or flexed knee gait, anterior tibial translation during weight-bearing, a delayed muscle activation, and altered recruitment order for relevant muscle groups.

There may be a process of relearning during neuromuscular training, where existing motor programs are adjusted to the patient's altered sensory condition^{76,81,103,110} in order to maintain within the envelope of homeostatic loading. This working hypothesis is further supported by the clinical observation that acutely-injured patients frequently report a poor control of extension (ie, the knee tends to go backwards), whereas chronic instability after an ACL injury is typically characterized by symptoms of giving way.

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